

SEDAR 12: Gulf of Mexico Red Grouper

Data Workshop Recommendations

Life History Group

- Studies performed with larger sample sizes for pre- and post-release mortality.
- All observer studies collect predation data and record release condition of fish.
- Future experimental studies to relate “sink or swim” observations to post-release mortality and suggests that controls are needed for all cage studies, such that control fish are captured and caged at depth (without bringing to the surface at all).
- Burns’ tag data be recoded to incorporate the comments regarding “sink or swim” into a standardized data field and used to estimate pre-release and predation mortality by sector.
- More research dedicated to determine methodologies to decrease release mortality (see Bartholomew and Bohnsack 2005).
- South Atlantic and Gulf Councils coordinate with CRP and MARFIN officers to provide all grant reports dealing with discards to be available at SEDARs and that all PI’s on grants dealing with said species are invited to SEDAR.
- All documents (including old assessments and references within) that were used in previous stock assessments for said species are more readily available to SEDAR participants.
- Conduct further review of current sampling methodologies by sector, including detailed comparison of length data from otolith samples and from more expansive port-based length sampling (via TIP; see SEDAR 12-DW-10).
- Bring increased attention to the need for strategies improving port sampling (representation of fishery sectors and random sampling)
- Increase the sampling of the recreational sector for biological samples throughout the docks and ports of Florida’s west coast.
- Continue support of fishery-independent surveys including all gears (hand-line, long-line, and trap) throughout the west Florida shelf.
- Continue exchanges of calibration otolith sets and age workshops among state and federal agencies and universities to continue improvements of data comparability and quality control.

- Continue use and development of a reference collection as a means to monitor precision between readers.
- Continue age structure sampling on an annual basis.
- Continue search for original samples and raw data on age and growth collected during the 1960s.
- Undertake more systematic collection of maturity data (e.g. to characterize the inshore and younger aged fish as well as the adults in mid and outer- shelf depths).
- Continue work on fecundity and spawning frequency and incorporate a spatial-temporal design to improve estimates of reproductive potential by age. Statistically test for regional effect. Continue work on spawning pattern to better understand and discriminate between annual asynchrony in spawning (skipped spawning) and seasonal asynchrony in spawning. Explore model sensitivities to reproductive parameters.

Commercial

- No research or monitoring recommendations provided.

Recreational

- Interviews/data on catch rates are needed from recreational fisheries prior to 1981, in order to improve estimates of historical catches.
- Study of discard mortality rates, preferably linked to factors that can be obtained from available recreational data.
- Discards undoubtedly have length/age frequency distributions which differ greatly from the landed catch, however there is little length or age information on these fish. Efforts should be made to collect such data. Collections methods could include length measurements of discarded fish obtained from anglers, at-sea observer programs, and/or the granting of special research permits allowing anglers to retain undersized fish as samples for researchers.

Indices Of Abundance

- The initiation and continued funding of such surveys, including, but not limited to the NE GULF INNER SHELF TRAP SURVEY. As trends can be regional in nature, the group highly recommends that recruitment trends be examined gulf-wide.
- Research be conducted to assess the possible impacts of hurricanes on the catch per unit effort of snapper/grouper complex members.

- Research be conducted to assess the possible impacts of red tide on the catch per unit effort of snapper/grouper complex members.

Assessment Workshop Recommendations

- Refine sampling for age determination to provide sufficient spatial and temporal coverage across all fisheries. Ensure some fisheries are not sampled excessively, necessitating subsampling for age determination.
- Quantify temporal and spatial changes in catchability rate
- Develop methods to evaluate the impact of natural events such as red tide in modeling M and the overall assessment.
- Develop and expand fishery-independent indices for tuning assessment models and evaluation of management measures
- Increase at-sea observation of discards by fishery to provide numbers of discards, fate of discards, and size/age composition of discards.
- Quantify release mortality rates by fishery by depth
- Improved the MRFSS survey and estimates of recreational fishing effort, especially to improve spatial resolution. Develop methods to obtain age samples from the recreational fishery and improve estimation of fish weight from recreational sampling.
- Support research to better describe and understand dolphin predation of red grouper.

Review Workshop Recommendations

Life History

- Investigate a two-gender growth model that explicitly addresses maturation and protogynous hermaphroditic gender change;
- Use tagging to further evaluate north-south connectivity;
- Explore temporal and/or density-dependent changes in growth and reproduction, including investigation of possible abiotic effects such as temperature;
- Publish a technical document about the application of Lorenzen method to convert conventional constant M to age-dependent M (avoid problem with the maximum age over which average has been developed).

Fishery

- Support ongoing work to evaluate and reduce possible bias and precision of recreational catch estimates;
- Evaluate sampling design for fishery length and age composition sampling for optimum cost, precision, analytical flexibility;
- Include more documentation of patterns in the fishery (seasonal, geographic, quota attainments, etc.) in the next assessment report.

Indices

- Evaluate the mix of surveys (longline, trap, SEAMAP video survey) to achieve best coverage of recruits and pre-recruits across relevant habitats and geographic and depth ranges.

Model

- Consider extending the model over different time periods. One sensitivity option would limit the assessment to the period after 1990 when the new 20 inch minimum size came into affect. Prior to 1990, data are different due to the size limit change so consider discarding pre-1990 data and fit the model to this shorter time series. Another option would be to complete the investigation of model performance and inference when the entire time series since 1880 is included. Such a long time series would have uncertainties due to assumptions about fishery characteristics in the early years, but could provide a check on the consistency between estimates of stock productivity and the cumulative removals over the entire time period.